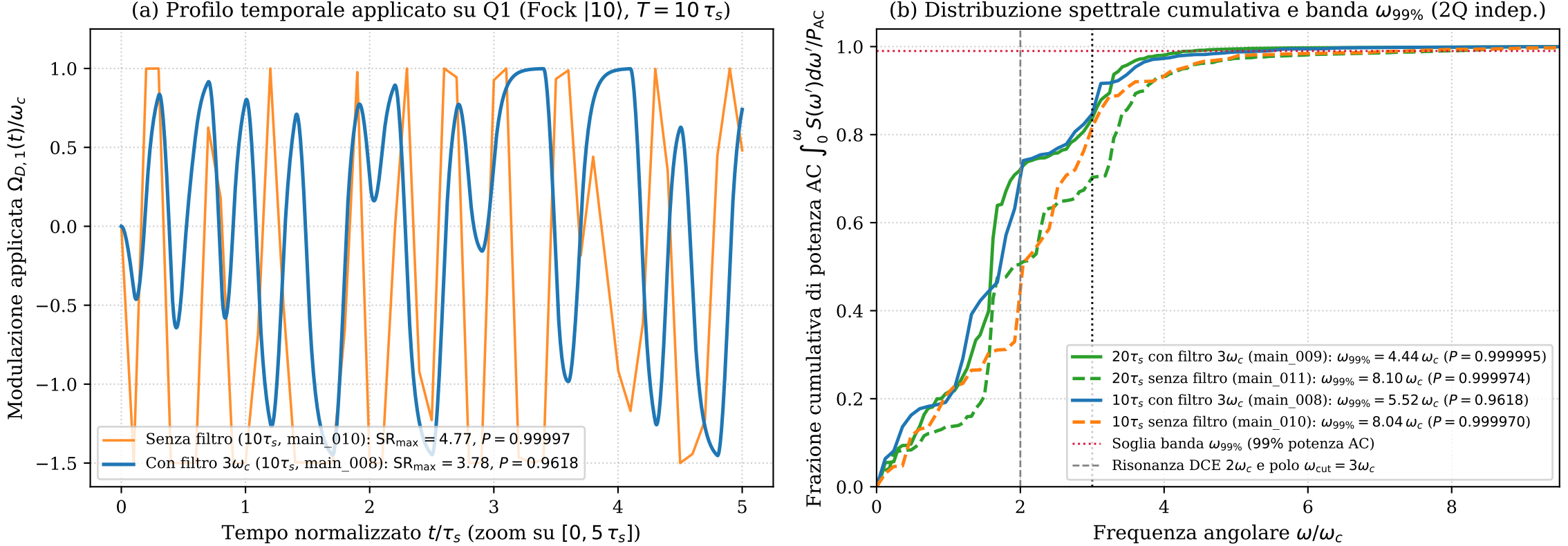
(a) Profilo temporale applicato su Q1 (Fock |10⟩, $T = 10\,\tau_s$)
Modulazione applicata $\Omega_{D,1}(t)/\omega_c$
Tempo normalizzato t/τ_s (zoom su $[0, 5\,\tau_s]$)
Senza filtro ($10\tau_s$, main_010): $SR_{max} = 4.77$, $P = 0.99997$
Con filtro $3\omega_c$ ($10\tau_s$, main_008): $SR_{max} = 3.78$, $P = 0.9618$
(b) Distribuzione spettrale cumulativa e banda $\omega_{99\%}$ (2Q indep.)
Frazione cumulativa di potenza AC $\int_0^\omega S(\omega')d\omega'/P_{AC}$
Frequenza angolare ω/ω_c
$20\tau_s$ con filtro $3\omega_c$ (main_009): $\omega_{99\%} = 4.44\,\omega_c$ ($P = 0.999995$)
$20\tau_s$ senza filtro (main_011): $\omega_{99\%} = 8.10\,\omega_c$ ($P = 0.999974$)
$10\tau_s$ con filtro $3\omega_c$ (main_008): $\omega_{99\%} = 5.52\,\omega_c$ ($P = 0.9618$)
$10\tau_s$ senza filtro (main_010): $\omega_{99\%} = 8.04\,\omega_c$ ($P = 0.999970$)
Soglia banda $\omega_{99\%}$ (99% potenza AC)
Risonanza DCE $2\omega_c$ e polo $\omega_{cut} = 3\omega_c$